

ARTICLE

Contagion or Confusion? Confounding Latent Embedding Adjustment for Networks (CLEAN)

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Abstract

Political phenomena often unfold in interconnected social networks, driven by two intertwined mechanisms: homophily (similar actors forming connections) and contagion (behavior spreading through connections). A core challenge in observational research is empirically disentangling these processes. Building on recent advances, we evaluate and extend a new method introduced by McFowland III and Shalizi (2023), which leverages **latent variable models** to separate contagion from homophily in network data. We refer to this approach as CLEAN—Confounding Latent Embedding Adjustment for Networks. Through extensive Monte Carlo simulations, we demonstrate CLEAN’s effectiveness in de-biasing contagion estimates. We also apply CLEAN to several influential studies in political network research to show its applicability. We conclude by discussing the broader implications and potential limitations of using CLEAN to study diffusion and interdependence in political networks.

Keywords: social networks, contagion, homophily

1. Introduction

Contagion is a core concept in political science. Researchers are interested in the diffusion of political attitudes, beliefs, and behaviors through social networks, such as voting preferences (Nickerson 2008; Bond et al. 2012), conflicts and wars (Buhaug and Gleditsch 2008; Danneman and Ritter 2014), and institutional changes (Linos 2011; Lankina, Libman, and Obydenkova 2016). Contagion studies also play a crucial role in causal inference, where researchers seek to address a common challenge to the stable unit treatment value (SUTVA) assumption (Egami 2021; Liu and Pang 2023). **It posits that the treatment of one actor should not affect another actor’s outcomes; yet, in reality, it often does.** The growing emphasis on social network research underscores the need for methods that account for and theorize these interdependencies among political actors’ treatments and outcomes.

Causally identifying contagious effects in observational data is notoriously difficult. The core challenge stems from the fact that contagion is often confounded by other mechanisms that drive connected units to exhibit similar attributes or behaviors (Franzese Jr, Hays, and Kachi 2012). Ogburn and VanderWeele (2014) summarize this problem as “inferring contagion in the presence of homophily.” *Contagion* refers to the social influence connected units have on each other, whereas *homophily* describes the tendency of similar nodes forming social ties between each other.¹ **Contagion and homophily often coexist, making isolating their effects extremely challenging.** Consider an example from the context of voter behavior: Does individual A turnout to vote due to the emulation of their friend, individual B,

1. Some scholars denote contagion as “social influence” and denote homophily as “selection.” To avoid confusion, we follow the convention by Shalizi and Thomas (2011) and use “contagion” as synonymous with social influence and “homophily” as synonymous with selection.

who always votes, or are A and B friends because they are both interested in politics, and as a result, both likely to vote (Bond et al. 2012)? When homophily follows unobserved/unmeasured attributes (i.e., is latent), contagion is highly challenging to identify (Shalizi and Thomas 2011).

To address the problem of latent homophily, McFowland III and Shalizi (2023) propose a method to separate contagious effects from latent homophilous effects in observational data. They propose to adjust estimates of contagion for latent variable embeddings of the network structure, using, specifically **stochastic block models (SBM) (Wang and Bickel 2017)** or **latent space models (LSM) (Hoff, Raftery, and Handcock 2002)**. This approach addresses a major limitation of the previous “Split-Halves (SH)” approach (Shalizi and Thomas 2011), which only identifies whether in general contagion exists in observed networks (Uppala and Desmarais 2023; Fyfe and Desmarais 2024). We term the approach proposed by McFowland III and Shalizi (2023) “confounding latent embedding adjustment for networks” (CLEAN). We evaluate CLEAN with a comprehensive simulation study and then apply it to three recent studies in political science, two of which are reported in the appendix.

2. The CLEAN Method

McFowland III and Shalizi (2023) use a causal diagram to represent **homophily as the confounder** when inferring contagion. When a network emerges through homophily, similar nodes are more likely to be connected with each other, resulting in clusters with dense within-group ties and sparse between-group links. However, in observational settings, homophily is often latent—researchers typically lack full information about traits that drive tie formation. As a result, failing to account for latent homophily can introduce omitted-variable bias when estimating contagion effects. To address this, McFowland III and Shalizi (2023) propose that in networks modeled with SBM or LSM, identifying and adjusting for latent variables provides a useful way to de-bias contagion estimates. We extend this approach to accommodate temporally varying adjacency matrices, which we summarize using the following regression model:

$$Y_{i,t+1} = \alpha_0 + \alpha_1 Y_{i,t} + \beta \frac{\sum_j (Y_{j,t} A_{ij,t})}{\sum_j A_{ij,t}} + \gamma_0^T \hat{C}_{i,t} + \eta_{i,t+1}, \quad (1)$$

$$\eta_{i,t+1} = \epsilon_{i,t+1} + \gamma_2^T X_i + (\gamma_1^T C_{i,t} - \gamma_0^T \hat{C}_{i,t}). \quad (2)$$

where Y is a random variable denoting the outcome of interest with n number of nodes $i \in \{1, \dots, n\}$ at time $t \in \{1, \dots, T\}$. The edges are observed and represented through an $n \times n$ adjacency matrix A which can vary over time. The time-invariant vector X_i represents the set of all other network-irrelevant attributes which affect $Y_{i,t}$ but not $A_{ij,t}$. The vector $C_{i,t}$ represents the true homophily for each i at time t . While $\hat{C}_{i,t}$, the recovered homophily through block detection, serves as a proxy to $C_{i,t}$. Intuitively, as a component of $\eta_{i,t+1}$, $\|\hat{C}_{i,t} - C_{i,t}\|$ is the statistical gap between the latent and true homophily and the detected homophily and plays a crucial role in CLEAN’s de-biasing performance. In the appendix (Section 1), we unpack the performance of CLEAN and confirm the importance of accurately identifying the latent variables that drive tie formation. A key takeaway is that CLEAN consistently performs better than an approach that ignores latent homophily, but the magnitude of improvements is determined by various features of the network and data generating process. We describe the comprehensive process below:

1. Given longitudinal network data with the adjacency matrix A , observe the temporal unit (e.g., yearly or monthly). Generate time-specific networks, represented as time-stamped adjacency matrices.
2. Apply block detection via the stochastic block model (SBM), to each adjacency matrix, resulting in a partition of the observations at time t

3. Incorporate a time-specific one-hot encoding of the detected blocks $\hat{C}_{i,t}$ into the regression model to obtain the de-biased coefficient β for the contagion variable $\frac{\sum_j (Y_{j,t} A_{ij,t})}{\sum_j A_{ij,t}}$.

3. Simulation Study

We design a Monte Carlo simulation study to evaluate the CLEAN method. We only focus on SBM-based networks because the estimation is more computationally efficient, and SBM can represent a broader variety of homophily structures. However, we expect a similar performance between SBMs and LSMs as suggested by McFowland III and Shalizi (2023). We describe our simulation setup below:

1. Begin with the number of nodes for the networks as $N \in \{50, 200\}$, the number of true clusters in the SBM-generated networks as $k \in \{2, 4\}$, and time period as $T \in \{15, 40\}$.
2. The true homophilous effect is set up as within-cluster probability $p_{\text{within}} \in \{0.25, 0.75\}$ and inter-cluster probability $p_{\text{between}} \in \{0.025, 0.10\}$.
3. The true block-specific effects are set up as $\gamma_1^T = [0.5 \quad -1 \quad -0.5 \quad 1]$, regarding the specific value of k . When $k = 2$, the first two values in γ_1^T are applied to $\mathbf{C}$, while when $k = 4$, this vector is fully applied to $\mathbf{C}$.
4. The true contagious effect is set up as $\beta \in \{0.1, 0.4\}$ for $\frac{\sum_j (Y_{j,t} A_{ij})}{\sum_j A_{ij}}$.
5. All the other time-invariant and network-invariant factors are wrapped up together, represented by a one-variable vector $\mathbf{X}_i$ with Gaussian distribution, with coefficient value $\gamma_2 \in \{0.1, 0.4\}$.
6. To induce variation in the magnitude of the block effects, we multiply γ_1^T by a constant $\tau \in \{0.1, 0.5\}$.

3.1 Results

There are in total 512 simulation conditions defined by the parameter grid. For each condition, we repeat the experiment for 100 iterations. In each iteration we run two regression models. The first model, *OLS*, contains all covariates, but excludes the detected block indicators $\hat{C}_{i,t}$. The second model, *CLEAN*, is identical to the first but includes the one-hot encoding of $\hat{C}_{i,t}$. To assess the performance of *CLEAN*, we focus on two metrics: bias and root mean squared error (RMSE). We expect *CLEAN* to return less-biased estimators for contagious effects. Variance is the other significant aspect of *CLEAN*'s performance, as *CLEAN* involves estimating more parameters than *OLS*. As such, it would be possible for *CLEAN* to be less biased but exhibit a higher RMSE than *OLS*. We therefore rely on RMSE as an integrated measure that covers both the performance of de-biasing and variability. In the appendix (Section 1.1), we also examine the coverage of the 95% confidence intervals. The results of our simulation study are presented in Figures 1–4.

The bias results are reported in Figure 1. We examine the degree to which the results vary based on the level of homophily (p_{within} and p_{between}), contagion (β), and block-specific (γ_1) effects. Figure 1 clearly shows that the *CLEAN* model yields smaller bias with the average of -0.027. Its distribution is noticeably close to a Gaussian, suggesting that *CLEAN* is a robust de-biasing method. In contrast, the *OLS* model shows greater bias, with an average of 0.267 and a right-skewed distribution. This comparison indicates that *CLEAN* outperforms the regular *OLS* in estimating the contagion estimator $\hat{\beta}$. In the appendix Section 1.2, we offer a comprehensive bias distribution among all the 512 groups of simulated networks.

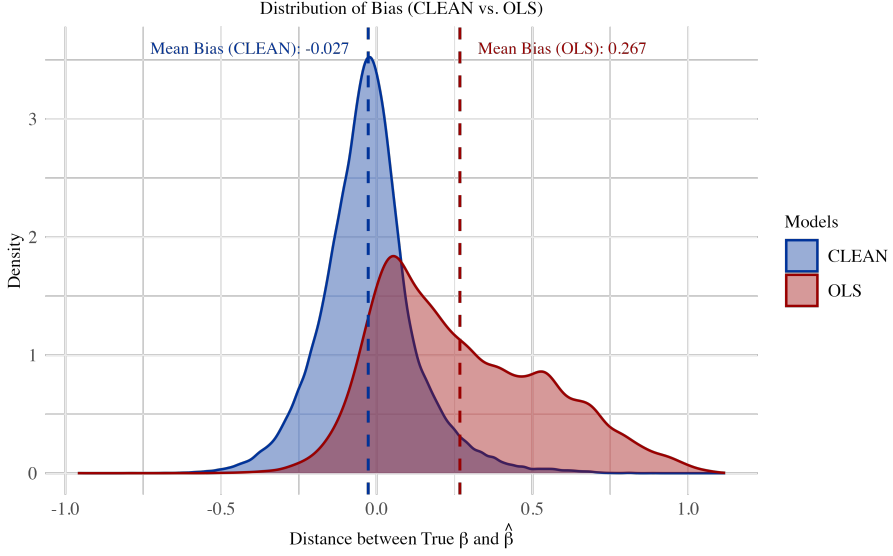


Figure 1. Comparison of Bias between CLEAN and OLS Models

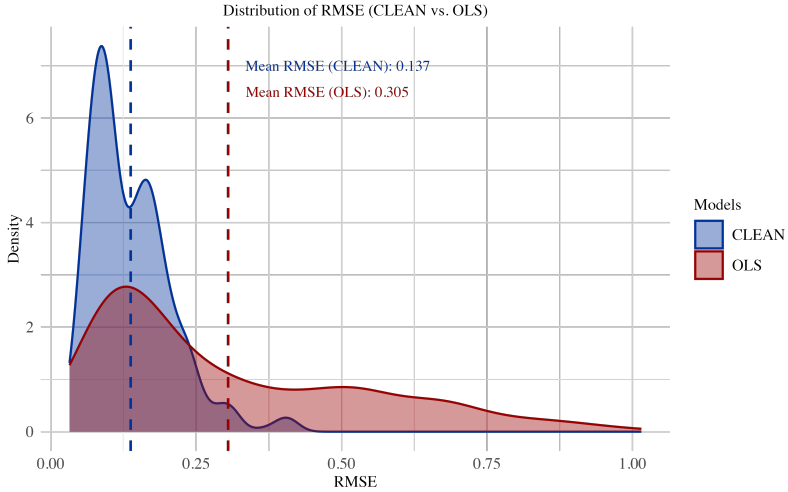


Figure 2. Comparison of RMSE between CLEAN and OLS Models

Second, we report results on RMSE. The density curves in Figure 2 reveal a stark contrast between the two models. CLEAN exhibits a distribution concentrated around a low RMSE value, with a mean of approximately 0.137. This indicates that CLEAN not only yields more precise estimates but does so consistently across various parametric configurations. In contrast, OLS displays a right-skewed distribution with a long tail, with a considerably higher mean RMSE of 0.305. This result indicates that CLEAN is more accurate overall, despite the estimation of additional parameters. In the appendix Section 1.2, we provide a more extensive presentation of the simulation results.

Third, to compare the performance of CLEAN and OLS across different simulation settings, we focus on the two most crucial parameters: the probability of within-block edge formation $p_{\text{within}} \in \{0.25, 0.75\}$ and the block-effect multiplier $\tau \in \{0.1, 0.5\}$. Both parameters are the principal

drivers of block detection accuracy under the SBM package in R.

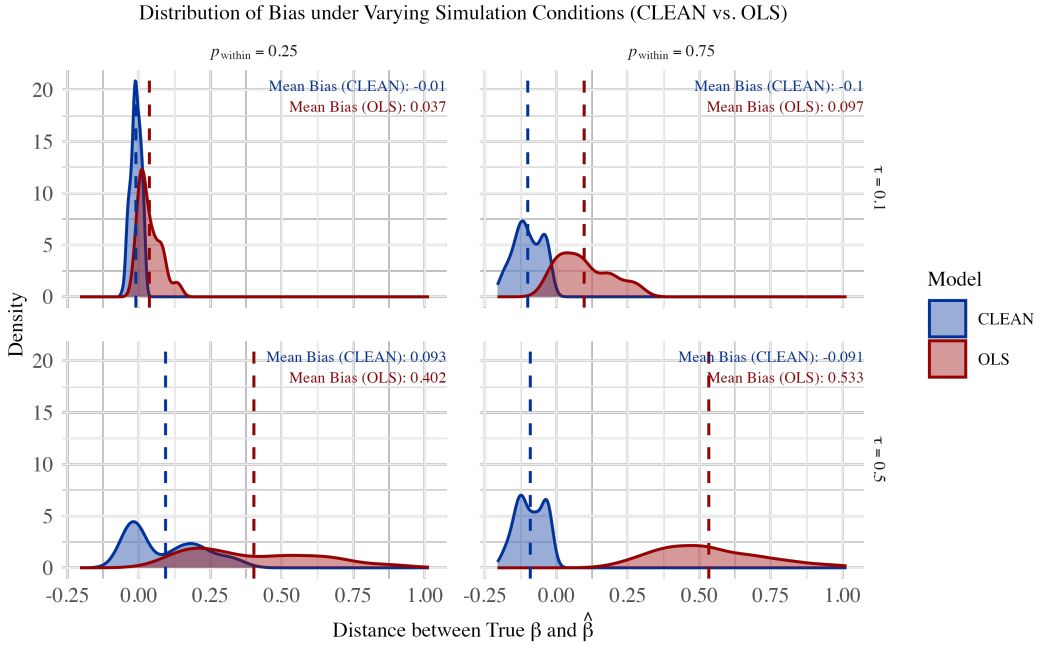


Figure 3. Performance of CLEAN and OLS Regarding Bias under Varying Simulation Conditions

Figure 3 displays the distribution of estimation bias for CLEAN and OLS across four sets of simulated networks (128 each) with varying p_{within} and τ . Across all conditions, CLEAN consistently demonstrates superior performance by yielding lower estimation bias with its distributions tightly concentrated around zero. **In contrast, OLS exhibits systematically larger biases, particularly when the block structure is more pronounced (higher τ) and within-block connections are stronger (higher p_{within}).** This is most evident in the lower-right panel, where OLS's bias sharply increases under strong block effects and high within-block density, while CLEAN remains comparatively stable. Figure 3 also shows that CLEAN's bias distributions are not only closer to zero on average, but also markedly less dispersed than those of OLS, indicating more reliable estimation. These findings reinforce that CLEAN effectively mitigates estimation bias arising from latent block structures, especially under conditions where OLS struggles due to unaccounted network dependencies.

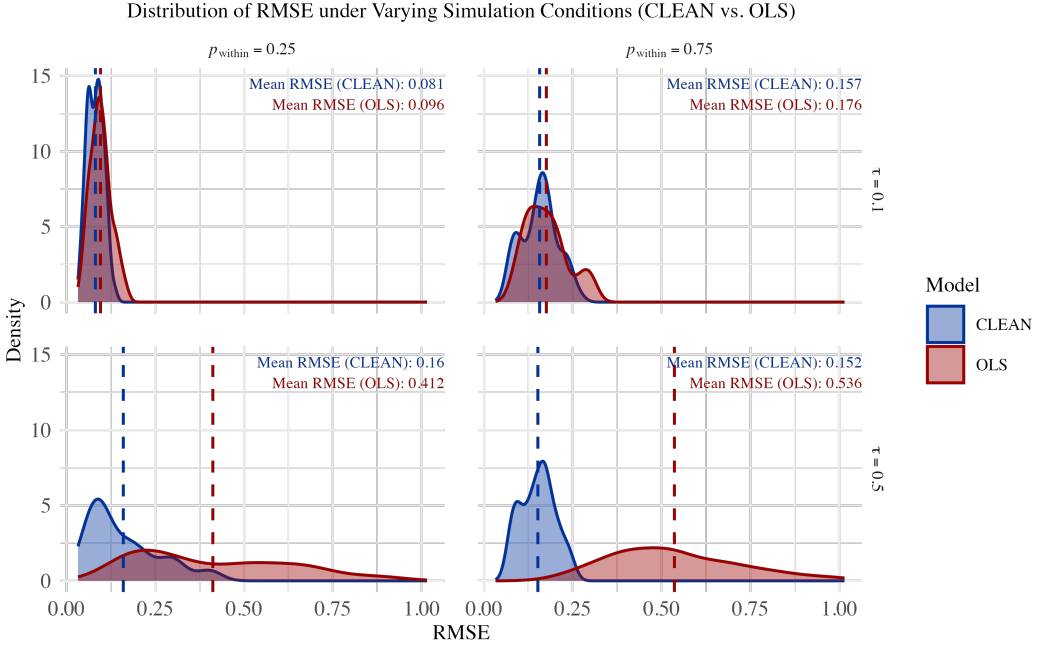


Figure 4. Performance of CLEAN and OLS Regarding RMSE under Varying Simulation Conditions

Figure 4 presents the distribution of RMSE for CLEAN and OLS under the same categories as in Figure 3. The results show that CLEAN generally outperforms OLS by achieving lower RMSE, even under scenarios with weaker block effects ($\tau = 0.1$). In the top-left and top-right panels, where τ is low, CLEAN consistently yields smaller RMSE values than OLS, with both models performing relatively well due to the modest structural complexity. However, as the block effect strengthens ($\tau = 0.5$, bottom panels), CLEAN maintains moderate RMSE levels, while OLS's RMSE inflates substantially, especially under high within-block connectivity ($p_{\text{within}} = 0.75$). This pattern highlights CLEAN's ability to effectively reduce estimation error in more complex network settings where traditional OLS fails to account for latent block structures.

4. Application: The Spread of State Repression

Recent studies in political science have highlighted the diffusion of state repression across regimes (Böhme, Ruggeri, and Pilster 2017; Olar 2019; Von Soest 2015; Mattes and Rodríguez 2014). We apply the CLEAN framework to replicate and extend the empirical analysis from Olar (2019), which offers both an ideal empirical structure for CLEAN, and strong justification for the use of SBM-based networks among authoritarian states. In the original analysis where the dependent variable is repression operationalized as a physical integrity index, two adjacency matrices are used: **one capturing institutional similarity based on ordinal regime-type classifications, and the other reflecting shared exposure to civil conflict, measured via annual civil war casualties**. We transform both general matrices into yearly adjacency matrices and detect latent blocks given each year with the `sbm` package in R. One-hot encodings of these yearly vectors of detected blocks are then reintegrated into the three core regression models constructed in the original analysis: one using only institutional similarity, one using only civil-conflict similarity, and one using both, the last of which is presented in Table 1.

Table 1. Comparison of Model 3 in Olar (2019)

	Original Model 3 Both peer groups	Model 3 with CLEAN Both peer groups
$W_{y^{\text{institutional similarity}}}$: ρ	0.028* (0.012)	0.027* (0.013)
$W_{y^{\text{experiential similarity}}}$: ρ	-0.001 (0.004)	-0.000 (0.004)
Lagged physical integrity index	0.901*** (0.012)	0.902*** (0.012)
Co-optation	-1.433*** (0.201)	0.573 (1.301)
Log GDP/capita_unit	-1.518*** (0.414)	1.251 (5.199)
Log population size_unit	0.195 (0.353)	1.559 (3.487)
Incidence of conflict	0.921*** (0.267)	1.124*** (0.330)
Share of conflict in neighbors	0.216 (0.728)	-0.031 (0.728)
Cold War	1.337 (1.243)	-0.655 (1.566)
Constant	19.487*** (5.600)	-22.796 (79.243)
Observations	3863.000	3863.000
R-squared	0.947	0.952
Standard errors in parentheses		
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$		

After incorporating the detected blocks, we find that the primary contagious coefficients of interest— ρ for $W_{y^{\text{institutional similarity}}}$ and $W_{y^{\text{experiential similarity}}}$ —remain stable in terms of sign, magnitude, standard errors, and statistical significance. Institutional similarity continues to exert a statistically significant influence, whereas experiential similarity does not. This supports the interpretation that autocrats tend to emulate repression strategies from institutionally similar peers, rather than from those merely experiencing similar civil conflicts and domestic casualties.

However, several control variables in Table 1 exhibit noticeable changes. For instance, the *Incidence of conflict* variable—measuring yearly casualty in each country—increases in magnitude while retaining its positive and significant effect, suggesting that the original analysis may have underestimated the domestic effect of civil conflict on autocrats' decisions to implement repressive strategies. Additionally, the variable *log GDP per capita* reverses its sign from negative to positive, but is only statistically insignificant. This sign reversal is both theoretically and empirically consequential. In the original analysis, higher levels of economic development were interpreted as a constraint on repression—implying that wealthier states, perhaps due to stronger institutions or greater public accountability, are less likely to rely on coercive strategies. However, once latent institutional clustering is accounted for through CLEAN, the opposite pattern emerges: economic development appears to correlate positively with the adoption of repression strategies. This shift suggests that the effect of wealth is not universal but may be conditioned by institutional context. Within certain types

of regimes—particularly those sharing institutional features with one another—greater economic resources may not inhibit but instead facilitate repressive actions. More specifically, increased state capacity could provide regimes with the means to monitor, suppress, and control dissent more efficiently, rather than serving as a liberalizing force.

These findings reinforce the central premise of CLEAN: omitting latent structural similarities—such as those captured by community detection algorithms—can lead to biased or incomplete conclusions about the behavior of covariates. By applying CLEAN, we not only recover latent block structures embedded in the observational data but also re-specify the relationships between contagion, homophily, and block-specific heterogeneity. In doing so, we separate these sources of dependence rather than allowing them to be conflated. The resulting shifts in key control variables, including *civil war* and *log GDP*, underscore how latent structures can condition the role and interpretation of otherwise well-established predictors. Rather than simply offering better model fit, CLEAN enables a more nuanced and structurally informed understanding of how repression diffuses and how its drivers operate differently across institutional environments. This structural sensitivity adds conceptual depth to existing theories of authoritarian behavior, showing that the dynamics of power and coercion are shaped not just by observable attributes, but also by hidden institutional alignments that shape learning, adaptation, and strategic emulation.

In Section 2 of the appendix, we provide more discussions on this application after applying CLEAN. We also present two other applications in the appendix, and discuss how the use of CLEAN affects the findings in those applications.

5. Discussion

The spread of political outcomes and behavior often reflects a combination of social influence and underlying similarity. Separating these processes in observational data remains a core challenge. Here, we propose a new approach that uses network clustering techniques to account for latent homophily. By adjusting for these unobserved structures, CLEAN allows for clearer identification of contagion effects.

Through simulations, we show that CLEAN reduces estimation bias compared to standard OLS. In an application to international repression diffusion, CLEAN preserves the key findings of the original study while revealing that certain covariates, such as GDP, change meaningfully when latent similarity is adjusted for. These shifts highlight the substantive importance of accounting for network confounding in political analysis.

CLEAN offers a flexible and accessible adjustment method for researchers working with network data. As research on diffusion and interdependence continues to grow, CLEAN provides a valuable tool for identifying contagion effects and improving inference.

Acknowledgments

We are grateful for insightful comments from audiences at PolNet 2024 and PolMeth 2024. We also thank the anonymous reviewers and editors at *Political Analysis* for their constructive feedback.

Competing Interests

Dr. Desmarais owns equity in Public Square Analytics, LLC. This financial interest has been reviewed by the University's Individual Conflict of Interest Committee and is currently being managed by the University.

Funding

This work was supported by the National Science Foundation [SES-2449569, BCS-2318460, SES-2148215] to Dr. Desmarais.

Data Availability Statement

All replication data and code used in this paper are available at the Dataverse repository: <https://doi.org/10.7910/DVN/XXXXXXX>. The simulation code used to generate synthetic network data is also archived on GitHub: <https://github.com/XXX>.

Supplementary Material

For supplementary material accompanying this paper, please visit XXX.

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